

Global Sales Business Intelligence

Dashboard

Tableau Dashboard Project

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Project Overview

This project presents an interactive Tableau dashboard analysing global sales performance from **2020-2023**. The dashboard provides business insights across sales, profit, products, customers, shipping performance, and profit forecasting to support data-driven decision making.

Dataset: A global retail sales dataset provided for academic analysis. The dataset is not redistributed in this repository.

Tools Used

- Tableau
- Business Intelligence
- Data Visualisation
- Forecasting
- Dashboard Design
- Data Analytics

Project Type

Portfolio Project

1. Executive Summary

This report summarises and analyses the business performance from 2020 - 2023

In this report, we would be analysing the company's performance in 5 dashboards.

Dashboard 1 – 5 will contain an overview of the company's performance, Customer Analysis, Product Analysis, Shipment Analysis and to predict business performance in the future, a forecast analysis will also be added.

Dashboard 1 shows the overall performance of the company including its top 3 markets namely USA, China and India in terms of profit while also showing its worst 3 performing markets which includes Turkey, Nigeria and Netherlands which shows losses. This shows which markets need intervention to improve performance.

Dashboard 2 focuses on product analysis which shows that Technology products such as phones and copiers contribute most to the profit of the company. We showed the top 3 products as well as the worst 3 products. This will help the business in what products need to be focused on to maximize profits and reduce losses. This dashboard also shows whether discounts influence our sales further helping to understand whether such discounts are needed to increase profitability.

Dashboard 3 focuses on our customers and their purchasing behaviour, retention rates and sales distribution among customer segments. By understanding when our customer taste and preferences and their purchasing habits, we will be able to optimize our business strategy and enhance customer loyalty to increase profitability in the long run.

Dashboard 4 provides insights on our shipping performance, allowing us to see what needs to be improved in terms of timely delivery. By optimizing our shipping deliveries and our operational efficiency, we will be able to achieve cost savings while enhancing customer satisfaction.

Finally, Dashboard 5 provides a forecast of future profits in different markets to know which markets need to be allocated more resources in order to improve profitability. This will ensure optimal resource allocation while maximizing returns which increases profitability for the business.

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2. Findings

2.1 Dashboard 1 - Overview Dashboard

This is an overview of the company's sales and profit performance along with key metrics. It shows the top 3 best and worst markets while showing an overview profit for each market.

Overview

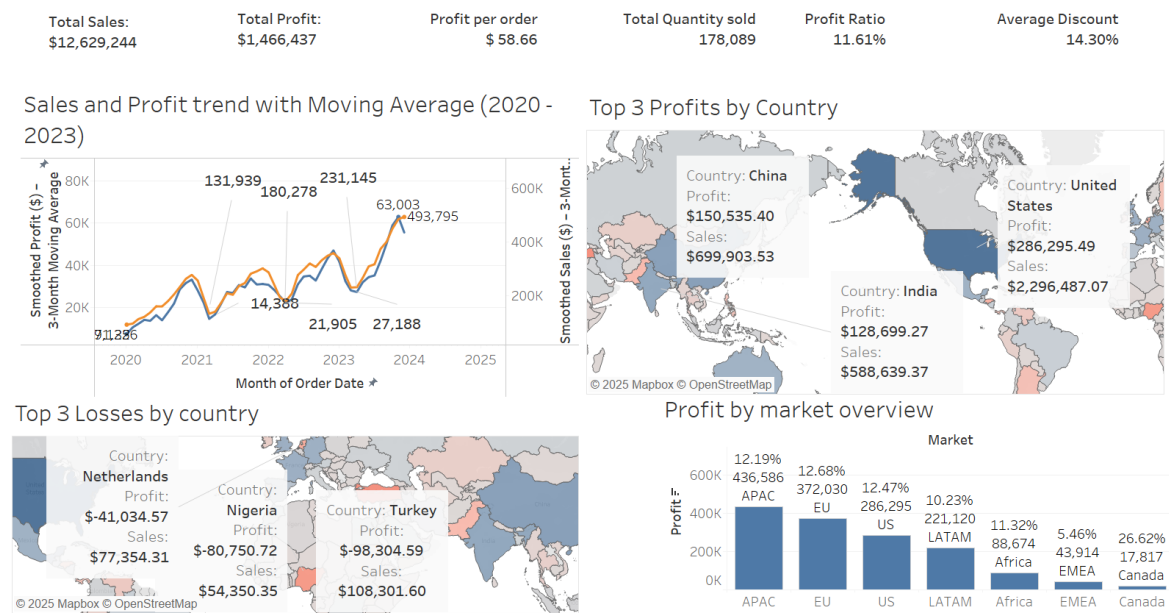


Figure 1: Overview Dashboard

The company generated \$12.629,244 in sales while having a profit of \$1,466,437. Figure 1 shows our top 3 countries in terms of profit. United States (\$286,295.49) , China (\$150,535.40) and India (\$128,699.27) respectively. This shows that these 3 markets are important in driving the profitability of the company. The sales and profit trend demonstrates a steady upward trend. We use moving average to help smooth out seasonal fluctuations and highlight periods of peak performance. From 2020 January profit of \$71,132 to 2023 December profit of \$55,602. This shows a significant increase in profit.

Profit by market graph shows that APAC, EU and the US are the strongest markets, while LATAM, Africa and EMEA shows lower profitability, indicating a need to focus on these markets.

This dashboard will help to show where the company excels and where improvements are needed. Thus, the business should expand on high profit markets while addressing loss making regions in order to maximize profitability.

2.2 Dashboard 2 – Product Analysis Dashboard

This dashboard focuses on the products the business makes including its top 3 performing and worst 3 products. In addition, this dashboard shows whether discounts influence sales and thus profitability.

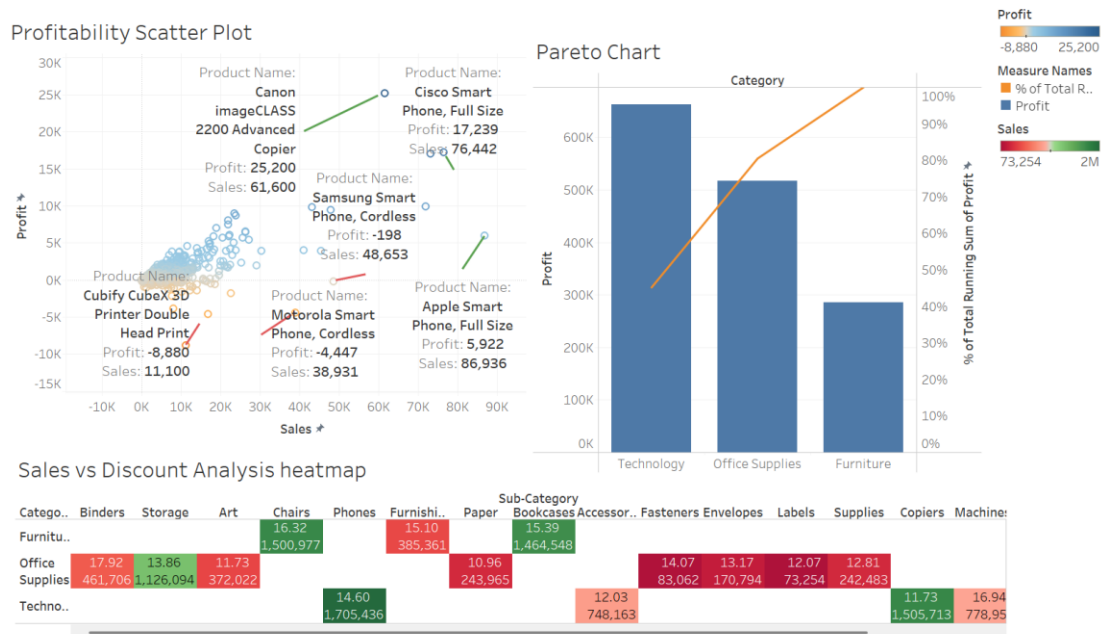


Figure 2: Product Analysis Dashboard

The profitability scatter plot shows all the products that the company delivered. The top 3 products by profitability are Canon ImageCLASS 2200 Advanced Copier (\$25200) in profit, Cisco Smart Phone full size (\$17239) and Apple Phone Full size (\$5922). It also shows the 3 worst performing products which include Samsung Smart Phone Cordless with (\$-198) in profit and Motorola Smart Phone Cordless (-\$4447) in profit.

Our Pareto Chart shows that technology products contribute the highest profits, followed by office supplies and then furniture. Technology and office furniture should be the key focus for the business profitability. The orange line represents the cumulative

percentage of total profit to 100%. There is a steep rise in Technology and Office Supplies. This implies that a few key categories namely in Technology and Office supplies contribute most of the profits.

The sales and discount analysis heatmap shows whether discount plays a significant role in increasing sales. Tables (29.04% discount, \$755,993 sales) and Machines (16.94% discount, \$778,953 sales) show high discount but moderate sales. Paper (10.96% discount, \$243,965 sales), Envelopes (13.17% discount, \$170,794 sales), and Labels (12.07% discount, \$73,254 sales) shows high discounts but low sales. This implies a weak correlation between discount and sales. While Storage (13.86% discount, \$1.1M sales) and Appliances (14.19% discount, \$1.01M sales) show high sales with moderate discounts. Thus, this heatmap shows that discounting does not strongly correlate with higher sales across all sub-categories. Thus, a targeted discount strategy is needed to maximize profitability.

2.3 Dashboard 3 – Customer Analysis Dashboard

This Customer Analysis Dashboard provides insights on customer retention, the top 10 customers, sales trends and customer segments. This dashboard will help us understand which customers drives sales and how it impacts business growth

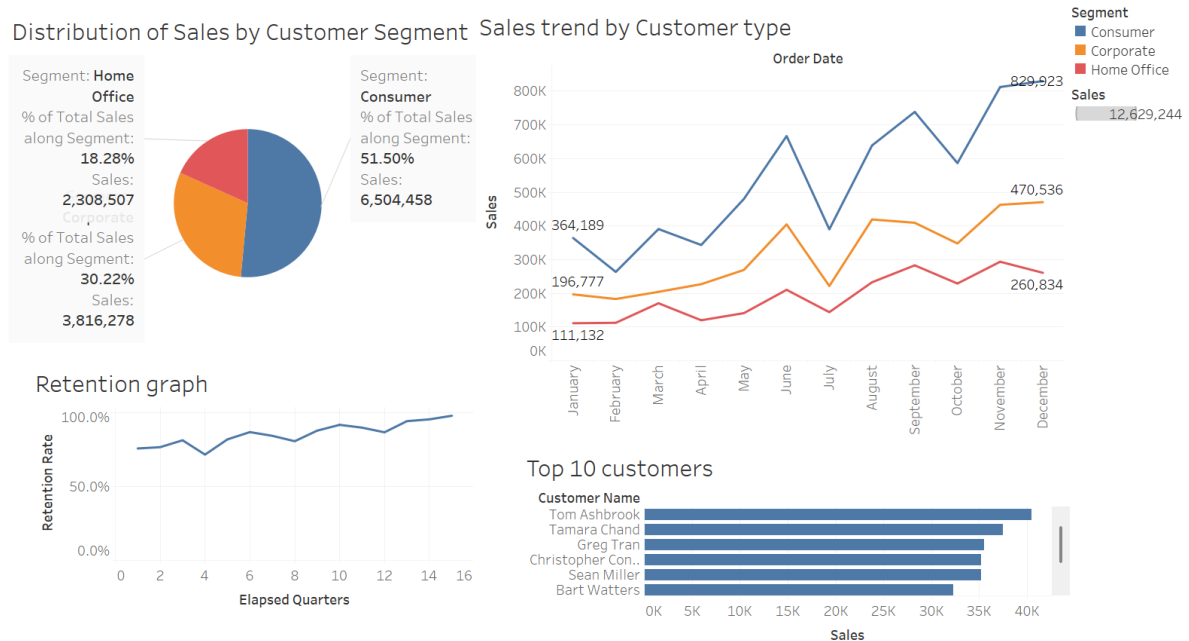


Figure 3: Product Analysis Dashboard

The consumer segment is the largest contributor of sales at 51.5% followed by Corporate at 30.22% followed by Home Office at 18.28%. This shows that consumer drives most sales however corporate can be looked at as a potential avenue for growth. Sales trend shows strong seasonality, with consumer sales peaking in December (\$829,923). Corporate sales also peaked in December (\$470,536) however growth remains steady throughout the year. Home office sales peaked in December (\$260,834) and shows consistent growth. The retention rate shows a strong retention rate of 80% indicating customer loyalty and repeat purchases. This can be further seen by our top 10 customers led by Tom Ashbrook, Tamara Chand, and Greg Tran contributing to most of our sales.

The business therefore should focus on Consumer and Corporate Segments while also leveraging seasonality for promotions by having targeted marketing efforts during December, April to June and July to August to maximize profitability.

2.4 Dashboard 4 – Shipment Analysis

This dashboard serves to provide insights into the timeliness of deliveries across different delivery methods, identifying areas where late shipments will affect customer satisfaction and loyalty which would impact the profitability and business operations.

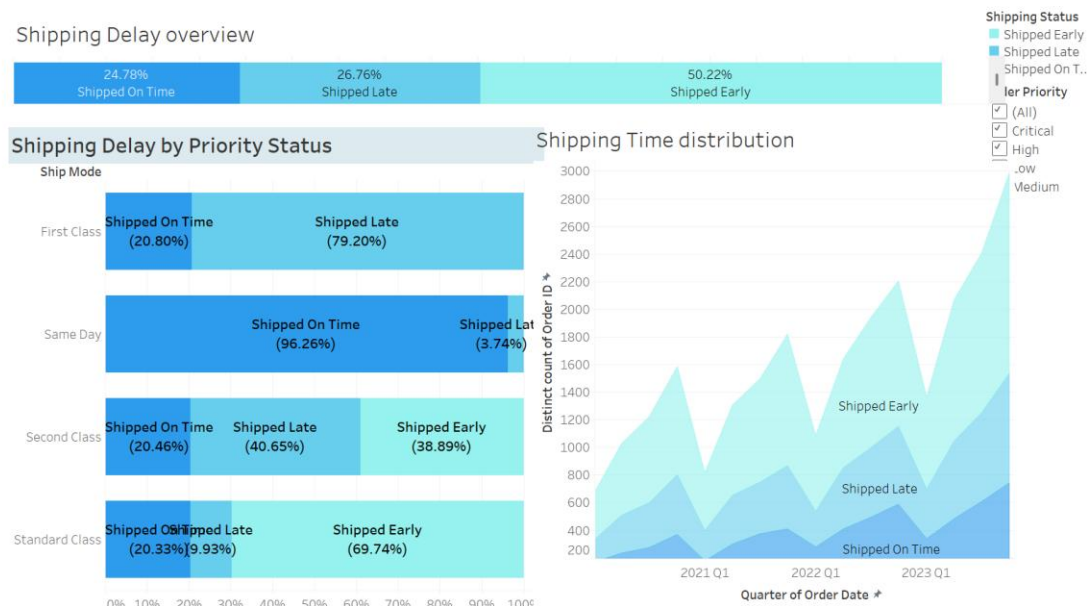


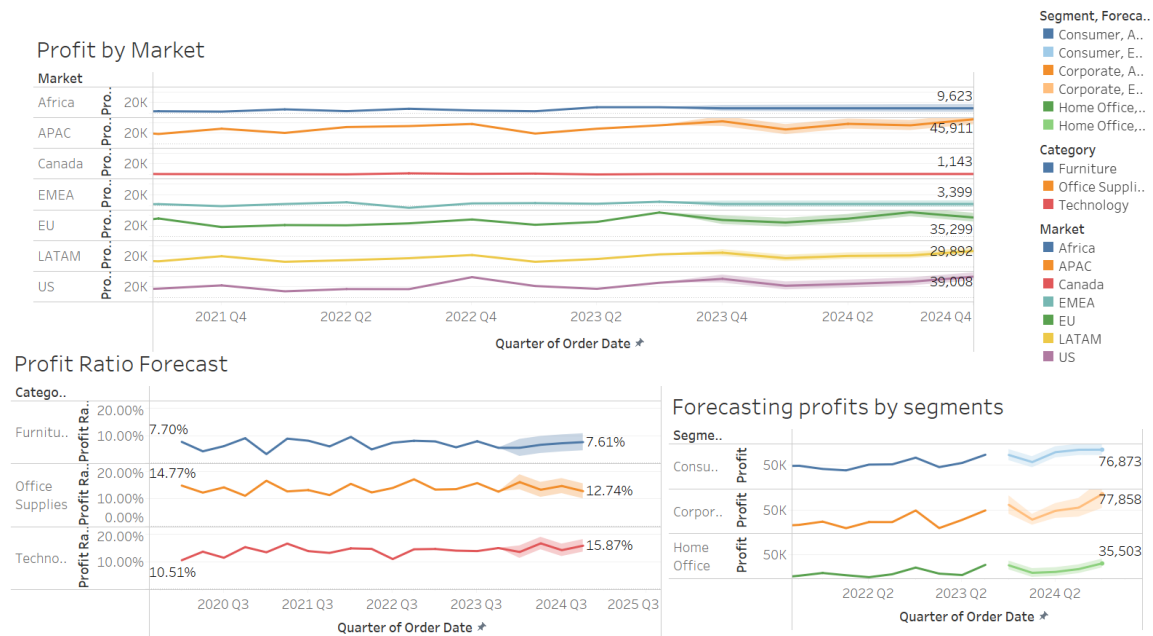
Figure 4: Shipment Analysis

An overview of our shipment delay shows that 50.22% of our products were shipped early while 24.78% were shipped on time. However, 26.76% of our products were shipped late. The products that came late was mostly shipped by First Class where 20.8% were shipped on time and 79.2% were shipped late. This indicates inefficiencies in our operation management for First Class orders and thus will need to be investigated. Same day shipping has a high reliability rate with 96.26%. Second class shipping has a 40.65% shipped late and this is concerning considered its expected reliability.

However, for the years 2020 – 2023 early shipments have increased over time which shows improvement in logistics however late shipments remain significant and have fluctuated which shows consistency challenges. To address this, we should focus on First Class and Second-Class shipping to address any issues that can lead to late shipments. Optimize standard class shipping as most standard class shipping comes early; costs can be reassessed to improve profit margins. Improving shipment reliability will increase customer satisfaction and loyalty. As well as operational efficiency which may lead to cost savings. Both measures can then maximize profitability for the business.

2.5 Dashboard 5 – Forecast of Profit

This dashboard provides information about the future profitability of markets for the business.



APAC (\$45,911) and EU (\$35,299) are projected to be strong contributors to the profit of the business. Office Supplies maintains the highest projected profit ratio (12.74%) which makes it a key area for revenue stability while technology (15.87%) shows steady growth which suggests strong market demand. However, Furniture (7.61%) has the lowest projected profit ratio and thus needs optimization and cost efficiency strategies.

Corporate consumers (\$76,858) are projected to contribute the highest profits while Home Office Customers (\$35,503) have the lowest projected profits thus requiring marketing strategies to attract more customers.

Thus we should focus on APAC, EU and US markets while optimizing our furniture category as it has the lowest profit ratio and thus need revaluation. We should leverage our Corporate Customers as they represent the most profitable segment.

3. Conclusion

In conclusion, these 5 dashboards provide a comprehensive understanding of our business performance. To enhance profitability, we can use Porter's 5 forces as a guide. We can strengthen our market position, optimize our product portfolio, enhance Customer Engagement, improve logistics efficiency and plan according to profitability potential by developing specialized services to our consumers. We should aim to focus on our strengths while addressing weaknesses, seizing opportunities and Mitigating threats.

The business can develop targeted marketing campaigns by using data on customer segments to create personalized marketing strategies that resonate with each group or implement loyalty programs to ensure high retention rate. To address the shipment delays we should address the delays in shipping while focusing our strengths on Standard-Class Shipping. As streamlining operations can lead to significant improvements in efficiency and profitability. For future operations by looking at the forecast dashboard we should invest in emerging markets such as APAC and EU while focusing on our corporate clients. This can be done by allocating more resources and offering specialized services. The Overview Dashboard shows us that we need to expand our high profit markets by increasing market share in United States, China and India which are the most profitable while addressing underperforming Regions like Netherlands, Nigeria and Turkey. All these measures will help the business in sustaining long term growth and profitability.

4. References

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